THE TOTAL CECH COMPLEX OF A BOUNDED COMPLEX OF FLAT SHEAVES IS K-FLAT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- In a distinguished triangle, if two of the three objects are K-flat complexes of modules then so is the third. invoked at proof lines 5916-5924. Runs the induction on the width of the bounded complex down to a single sheaf.
- The Cech complex of a covering agrees up to homotopy equivalence with the alternating Cech complex. invoked at proof lines 5926-5931. Passes from the total Cech complex to the alternating one whose terms are sections over the intersections.
- A bounded complex of flat modules is K-flat. invoked at proof lines 5936-5940. Concludes the proof once the terms are seen to be A-flat.

Source reference. Original lemma 06Y2: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5916-5924 to run the width induction; the triangle it is applied to is set up inside the counted span.

Source reference. Original lemma 01FM: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5926-5931 to pass to the alternating complex; the flatness argument on its terms is inside the counted span.

Source reference. Original lemma 064K: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5936-5940 to conclude; the verification that the terms are flat over A is inside the counted span.

Lemma 0.2. *Let $f : X \to Y$ be a morphism of schemes with $Y = \mathrm{Spec}(A)$ affine. Let $\mathcal{U} : X = \bigcup_{i \in I} U_i$ be a finite affine open covering such that all the finite intersections $U_{i_0 \ldots i_p} = U_{i_0} \cap \ldots \cap U_{i_p}$ are affine. Let $\mathcal{F}^\bullet$ be a bounded complex of $f^{-1}\mathcal{O}_Y$-modules. Assume for all $n \in \mathbf{Z}$ the sheaf $\mathcal{F}^n$ is a flat $f^{-1}\mathcal{O}_Y$-module and $\mathcal{F}^n$ has the structure of a quasi-coherent $\mathcal{O}_X$-module compatible with the given $p^{-1}\mathcal{O}_Y$-module structure (but the differentials in the complex $\mathcal{F}^\bullet$ need not be $\mathcal{O}_X$-linear). Then the complex $Tot(\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}^\bullet))$ is K-flat as a complex of A-modules.*

Proof. We may write

$$\mathcal{F}^\bullet = (\mathcal{F}^a \to \ldots \to \mathcal{F}^b)$$

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Arguing by induction on $b - a$ and considering the distinguished triangle

$$\mathcal{F}^b[-b] \to (\mathcal{F}^a \to \ldots \to \mathcal{F}^b) \to (\mathcal{F}^a \to \ldots \to \mathcal{F}^{b-1}) \to \mathcal{F}^b[-b+1]$$

and using More on Algebra, Lemma 06Y2 we reduce to the case where $\mathcal{F}^\bullet$ consists of a single quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ placed in degree 0. In this case the Čech complex for $\mathcal{F}$ and $\mathcal{U}$ is homotopy equivalent to the alternating Čech complex, see Cohomology, Lemma 01FM. Since $U_{i_0 \ldots i_p}$ is always affine, we see that $\mathcal{F}(U_{i_0 \ldots i_p})$ is A-flat. Hence $\check{\mathcal{C}}^\bullet_{alt}(\mathcal{U}, \mathcal{F})$ is a bounded complex of flat A-modules and hence K-flat by More on Algebra, Lemma 064K. $\square$